

Pablo Pena

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EDUCATION

University of Texas at Austin

B.S. in Mechanical Engineering Honors, Robotics Minor

August 2024 – Fall 2027

GPA: 4.0

SKILLS

CAD & Analysis: SolidWorks, Siemens NX, FEA, GD&T

Manufacturing: 3D Printing, Laser Cutting, SMT Soldering, Wire Harness Fabrication, CF Layups

Programming: Python, MATLAB, C#, Arduino, SQL, Java

PROFESSIONAL EXPERIENCE

BlackPearl Technology

June 2026 - August 2026

Mechanical Engineering Intern

- Reduced component count by 67% through design consolidation, streamlining manufacturability and reducing assembly complexity
- Reduced product assembly times by 60% through fixture and process improvements
- Designed and documented 30+ components and assemblies for production and testing applications using SolidWorks
- Managed a 100+ line-item BOM by implementing engineering changes, including vendor updates, part revisions, hardware selection, and new component integration to support purchasing and inventory

Landry's Inc.

June 2024 - August 2024

Information Technology Intern

- Developed a Power BI dashboard consolidating 9 key IT performance metrics, improving visibility and streamlining reporting across departments
- Reviewed and corrected 180+ intranet entries to improve data accuracy and maintain up-to-date operational information across restaurant locations
- Developed a C# application leveraging APIs to automate data transfers between loyalty systems, eliminating manual CSV processing

PROJECTS

RC Plane — Texas Design Build Fly

August 2024 - Present

- Designed and fabricated 4 fuselage prototypes using hybrid composite layups and vacuum bagging, optimizing stiffness-to-weight ratio and manufacturability
- Manufactured landing gear with carbon fiber/Kevlar with $\pm 45^\circ$ CF plies for increased torsional stiffness and 0° Kevlar plies for deflection resistance; validated layup quality through fit checks and assembly testing

Battle Bot

January 2025 - May 2025

- Designed and fabricated 10 custom components using SolidWorks, 3D printing, and laser cutting to create a sub-1 lb combat robot
- Selected and integrated drive, actuation, and power subsystems, including DC motors, servo actuation, compression springs, and a 7.4 V LiPo battery pack, to improve durability and weapon performance

ACHIEVEMENTS & AWARDS

- 6th Place Overall — AIAA Design/Build/Fly International Competition

April 2025

- Hispanic Scholarship Fund (HSF) Scholar

December 2024